

Thrust Measurement System: HX711 & 20kg Load Cell

This manual outlines the setup and operation of the high-precision thrust measurement device, utilizing a 20 kg load cell and the HX711 24-bit ADC module interfaced with an Arduino microcontroller.

Technical Specifications

- Load Capacity: 20 kg Maximum
- ADC Resolution: 24-bit (HX711)
- Input Voltage: 2.7V – 5V DC
- Interface: Serial (Data and Clock)

Hardware Integration

The system requires four primary connections between the HX711 amplifier and the Arduino board. Ensure all connections are secure and the load cell is mounted on a rigid base for accurate readings.

HX711 Pin	Arduino Pin	Description
VCC	5V	Power Supply
GND	GND	Ground Reference
DT (Data)	D2	Digital Data Pin
SCK (Clock)	D3	Serial Clock Pin

How to use the device

1. Initial Calibration

After connecting the device, wait for it to calibrate.

2. Zeroing (Taring)

To tare the device, press the 't' key (case-insensitive) on your keyboard while measuring weight and when the initial calibration has been completed. Ensure the load cell is free of any unintended weight or mechanical stress during this phase to establish a clean zero-point.

3. Measurement Procedure

Once tared, apply thrust or weight to the load cell. The Arduino will process the differential voltage via the HX711 and output the thrust value through the Serial Monitor or an attached display. To view thrust measurements, open your saved 'thrust measurement' profile in PuTTY. Ensure you have selected the appropriate COM port and set the baud rate to 9600.

Maintenance & Safety

- **Overload Protection:** Do not exceed the 20 kg limit. Physical deformation of the load cell can lead to permanent sensor damage and inaccurate data.
- **Vibration Control:** For thrust measurement (e.g., motor testing), ensure the mounting bracket is damped to minimize high-frequency noise in the readings.
- **Wiring Integrity:** Periodically check the solder joints at D2 and D3, as vibrations from thrust testing can loosen mechanical connections.